RoboPIN: Grounded Embodied Reasoning via Pinned Chain-of-Thought

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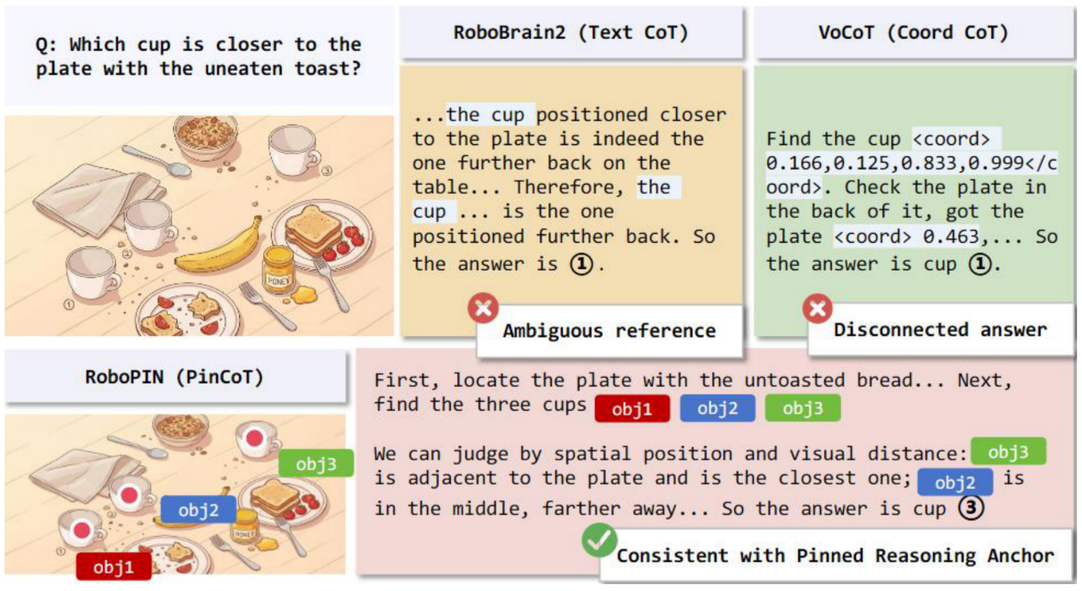
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Motivation



1. Text-based references can drift between similar objects. Coordinate-only grounding lacks explicit identity tracking.

2. A model may correctly analyze object A, then silently point to object B. This reasoning-to-answer misalignment breaks reliable visual grounding.

Key Insight: Persistent Identity

Implicit references

a cup → it → the target

Which instance is carried forward?

PinCoT: persistent identity

obj_01 → obj_01 → obj_01

- **Spatial references are usually isolated annotations, without explicit identity tracking during reasoning.**
- **PinCoT binds each entity to a name, ID, view and point. Later steps reuse the same ID.**
- **Our insight: visual anchors act as persistent reasoning variables, making each step traceable to visual evidence.**

QR PLACEHOLDER

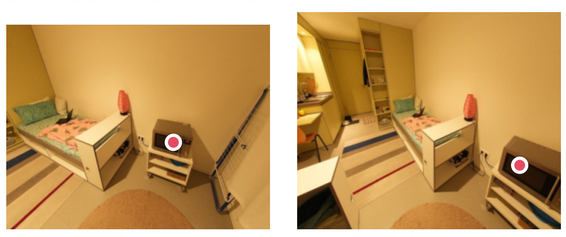
PAPER / CODE

Our Method: RoboPIN / PinCoT

A. How PinCoT Reasons

1. Observe

How did the camera rotate between the two views?



View 0 View 1

2. Introduce an anchor

```
<obj name="microwave"
id="obj_01" img_idx="0"
point="[634,576]">
```

Introduce the entity once with identity, view and point. Then refer to it by obj_01.

3. Reuse identity

Same obj_01, new view:

View 0: [634,576]
View 1: [844,706]

Its x-coordinate increases from 634 to 844: the object appears to shift right.

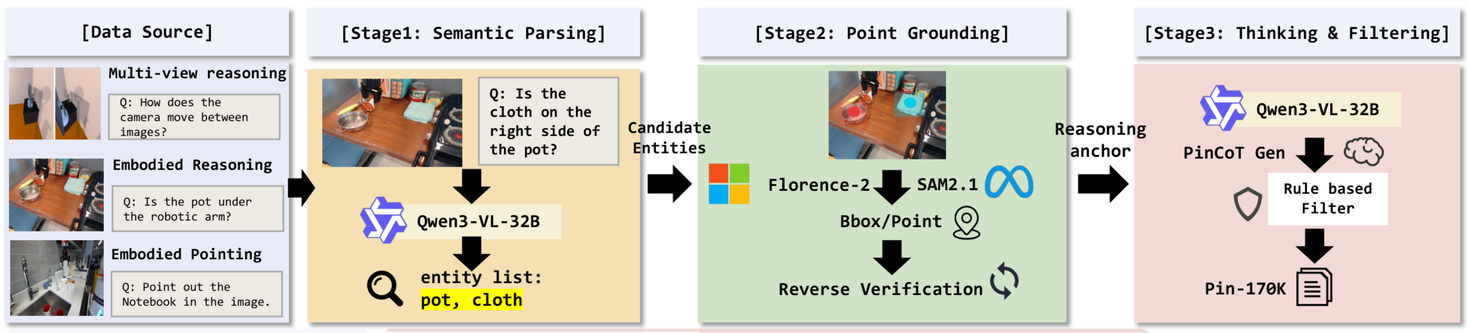
4. Ground the answer

In this example, apparent object motion is opposite to camera motion.

Answer

Movement left

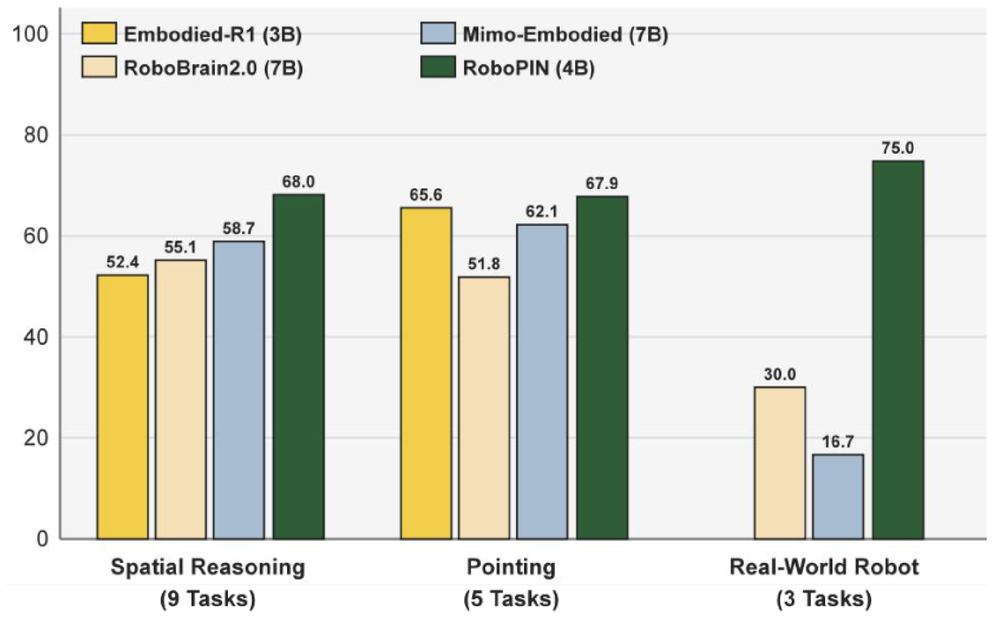
B. Data Construction & Training



Learning to reason with anchors

SFT	Embodied knowledge
CoT-SFT	Learn PinCoT traces
RFT / GRPO	Align reasoning and answers

Experimental Results



Ablation studies

Reward design

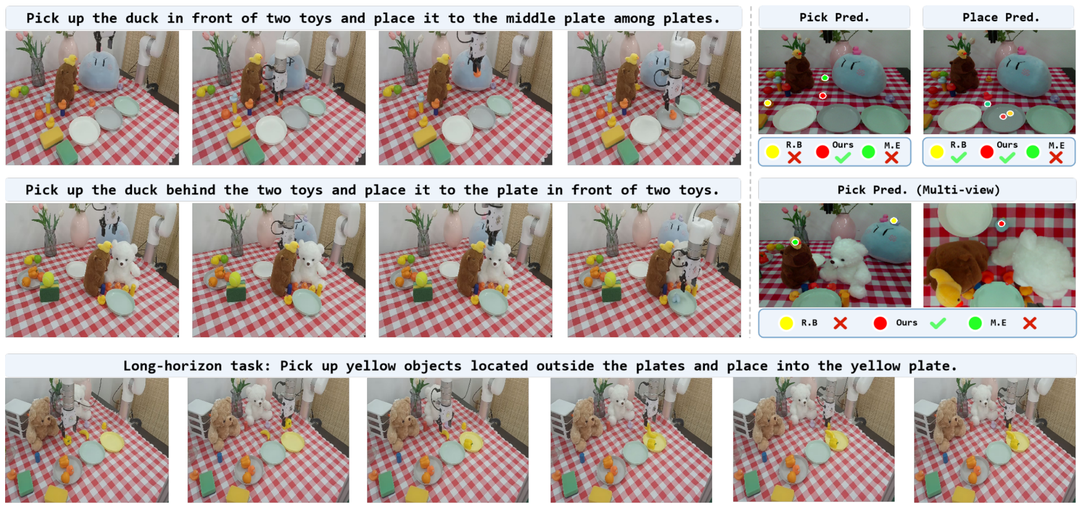
Setting	EmbSpatial
Full reward	**84.7**
w/o Pin	82.0
w/o Consistency	82.7

Training stages

Stage	Spatial	Point
SFT	57.0	66.0
+ CoT-SFT	62.0	63.6
+ RFT	**67.7**	**67.9**

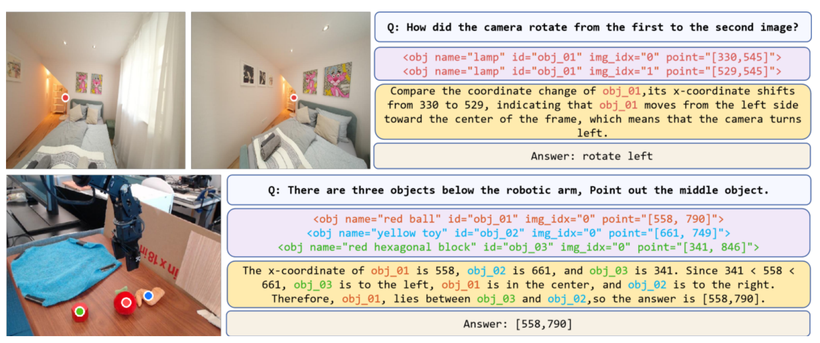
Spatial reasoning: +10.7 points from SFT to RFT.

Qualitative Results



Zero-shot xArm · successes / 20

20 / 20	14 / 20	11 / 20
Spatial	Multi-view	Long-horizon



Persistent Visual Anchors · Consistent Reasoning · Grounded Answers